14.3. Irradiation of ^{238}U with deuterium yields ^{238}Pu. Will any other plutonium isotope be produced?

14.4. What are the decay products of ^{252}Cf, of ^{253}Cf and of ^{254}Cf?

14.5. ^{244}Pu decays through spontaneous fission with a half-life of 6.6×10^{10} y. Estimate the number of neutrons emitted per fission if the measured n-emission rate is 1890 n s^{-1} g^{-1}.

14.6. Flerov bombarded ^{207}Pb with ^{54}Cr and obtained a product which within 4-10 ms decayed by spontaneous fission. Suggest a product nucleus.

14.7. In Figure 14.9, Eu(III) is more easily extracted than its homolog Am(III). Suggest an explanation.

14.8. A cardiac pacemaker contained 150 mg of ^{238}Pu. What is its heat output? Use data in the isotope chart.

14.9. What electronic configuration would you ascribe to nobelium considering its place in the periodic system and its chemistry?

14.10. Based on the redox diagrams in Figure 14.8, estimate the reduction potentials for the one-electron step in the following series of plutonium species in acid solution: Pu(III), Pu(IV), Pu(V), and Pu(VI).

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CHAPTER 15

Radiation Biology and Radiation Protection

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